

EDWARD M. LIU

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SUMMARY

Senior machine learning engineer with 10+ years of experience designing, building, and deploying production ML and AI systems for health and consumer products. Deep expertise in generative AI, NLP, and deep learning applied to clinical and biomedical workflows. Proven cross-functional leader: scoping end-to-end ML pipelines, shipping customer-facing features, and building evaluation frameworks and human-in-the-loop review that earn trust. Experienced with healthcare data (EHR, genomics, clinical notes) and modern agentic AI tooling (prompt engineering, RAG, tool use, multi-agent systems) to scale intelligent solutions across teams and institutions.

EXPERIENCE

Verily Life Sciences

Staff Data Scientist

2025 - Current

Mountain View, CA

- Delivered an end-to-end clinical NLP system from zero to production-ready in four weeks; built a generative AI-powered abstraction pipeline for unstructured molecular reports with evaluation suite and reviewer workflow, achieving > 95% accuracy and unlocking Verily's largest commercial deal.
- Scaled applied LLM development across real-world healthcare datasets by automating prompt optimization with DSPy and establishing repeatable offline evaluation and benchmarking frameworks.
- Co-authored multimodal EHR foundation model research (2nd author); transformed an inherited R&D codebase into an open-source ETL and ML stack; now leading v2 model R&D integrating unstructured EHR data with deep learning.

Paradigm Health

Staff Machine Learning Engineer

2023 - 2025

New York, NY

- Led a team of four ML engineers building production LLM, NLP, and deep learning systems integrating structured and unstructured EHR data into clinical workflows for patient-trial matching.
- Designed and deployed an LLM-driven clinical extraction system, improving F1 by 10 points; implemented failure analysis, golden datasets, and regression testing for production stability and auditability.
- Collaborated cross-functionally with product, clinical, and engineering teams to ship new ML models across data platforms and customer-facing products.
- Embedded with partner clinical ops teams to design and deploy human-in-the-loop validation into production abstraction workflows, balancing automation with clinician trust.
- Prototyped retrieval-augmented generation (RAG) and agentic patient-trial matching capabilities with tool grounding, explainability, and multi-step reasoning.

Flatiron Health

Senior Data Scientist (previously Senior SWE, SWE)

2019 - 2023

New York, NY

- Shipped ML-powered systems combining structured EHR data, clinical notes, and genomics to identify trial-eligible cancer patients, achieving 3× expected enrollment—directly impacting customer-facing products.
- Built production data pipelines and web application features for genomic data extraction and integration, powering complex clinical trial workflows and analytics at scale.
- Trained novel deep learning models to predict cancer treatment response from unstructured data, resulting in new commercial real-world data products.

TripleLift

Data Scientist

2016 - 2018

New York, NY

- Designed and deployed predictive models for real-time ad performance, improving click-through-rates by 22%.
- Led A/B experimentation and optimization initiatives, increasing advertiser spend by 12%.

EDUCATION

Columbia University

M.S. in Computer Science (Machine Learning), GPA: 3.8.

B.S. in Computer Science, Minor in Applied Mathematics

Teaching Assistant for undergraduate and graduate CS courses; Master's supported by TA Fellowship

New York, NY

2016

2014

SKILLS

GenAI & ML Engineering	Prompt engineering, structured outputs, tool use, RAG, agentic systems (routing, planning, multi-agent, memory management), LLM evaluation and benchmarks, HITL review
Deep Learning & NLP	PyTorch, Hugging Face Transformers, DSPy, LangChain, LlamaIndex, Scikit-learn, NLTK, John Snow Labs, NVIDIA NeMo
Data & Infrastructure	Spark, Databricks, Pandas, Airflow, dbt, Snowflake, MLflow
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, Terraform
Programming	Python, SQL, Java
Leadership	Team leadership, cross-functional collaboration, scientific communication, rapid prototyping in fast-paced environments

PUBLICATIONS & PATENTS

Amar, J., **Liu, E.**, et al. “Integrating Genomics into Multimodal EHR Foundation Models.” *arXiv:2510.23639*, 2025. [Link](#).

Lakhanpal, S., et al. “An Automated EHR-Based Tool to Facilitate Patient Identification for Biomarker-Driven Trials.” *Journal of Clinical Oncology*, 2021.

Co-inventor on three healthcare-focused machine learning patents (trial matching, genomic extraction, response-to-treatment modeling).